

SHANGHAI PUDONG INTL, China

WMO: 583211

Lat: 31.143N

Lon: 121.805E

Elev: 4

StdP: 101.28

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.2	-1.0	-12.9	1.2	2.9	-10.0	1.6	3.2	11.5	2.8	10.4	4.3	3.6	300	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.4	35.1	27.3	33.8	27.1	32.2	26.6	28.3	32.3	27.7	31.3	27.2	30.6	5.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.2	22.9	30.5	26.9	22.6	30.2	26.2	21.6	29.6	90.8	32.8	88.2	32.1	86.3	30.8	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.0	9.7	8.8	-4.4	37.5	1.7	1.5	-5.6	38.6	-6.6	39.4	-7.6	40.3	-8.9	41.3
			-5.9	29.4	1.9	0.6	-7.3	29.8	-8.4	30.2	-9.5	30.6	-10.9	31.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.2	5.0	6.2	10.2	15.5	20.5	24.0	28.9	28.7	24.8	20.0	14.2	7.6
	DBStd	8.75	3.19	3.36	3.55	3.27	2.51	2.57	2.41	1.98	2.20	2.54	3.65	3.61
	HDD10.0	414	159	113	42	2	0	0	0	0	0	0	8	91
	HDD18.3	1587	414	339	254	98	8	0	0	0	0	12	130	332
	CDD10.0	3038	4	8	47	165	326	421	584	579	446	309	134	17
	CDD18.3	1169	0	0	1	12	75	171	326	321	196	63	6	0
	CDH23.3	10569	0	1	8	73	347	1002	3867	3774	1336	152	10	0
	CDH26.7	3915	0	0	1	15	78	293	1704	1546	265	12	0	0
Wind	WSAvg	4.7	4.7	4.8	4.9	4.8	4.7	4.3	4.7	5.0	4.9	4.7	4.5	4.6
Precipitation	PrecAvg	1264	69	70	80	91	100	211	135	198	118	72	67	52
	PrecMax	1686	200	161	186	160	220	687	242	401	267	317	133	118
	PrecMin	847	13	18	29	38	34	25	54	13	18	6	8	0
	PrecStd	225	43	38	41	33	46	150	56	99	56	77	41	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.8	19.2	23.8	28.9	31.2	34.2	37.2	37.0	32.2	28.1	24.8	19.0
		MCWB	12.0	14.6	16.6	19.0	21.1	26.7	27.3	27.7	25.7	24.1	19.1	15.2
	2%	DB	13.2	15.9	20.1	25.2	28.9	32.1	35.9	34.2	30.8	26.1	22.2	16.2
		MCWB	10.3	12.3	14.3	18.4	20.3	25.8	27.1	27.3	25.8	22.1	18.3	12.6
	5%	DB	11.2	13.1	18.0	22.9	26.9	30.0	34.2	33.1	29.1	25.0	21.0	14.2
		MCWB	8.8	10.0	12.9	17.2	20.0	24.9	27.2	27.1	24.9	20.7	17.8	11.5
	10%	DB	9.8	11.1	15.9	20.9	25.1	28.0	33.0	32.0	28.1	23.9	19.8	13.0
		MCWB	7.8	8.6	12.4	16.7	19.3	24.2	27.1	26.9	24.2	19.7	16.8	10.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.6	15.5	17.5	21.6	24.4	27.6	29.3	29.1	27.1	25.5	21.5	17.2
		MCDB	15.7	18.5	21.3	25.6	28.7	32.4	34.5	33.8	29.5	27.5	23.6	18.4
	2%	WB	11.2	12.9	15.6	19.7	22.7	26.6	28.5	28.3	26.5	23.7	20.3	13.9
		MCDB	12.6	15.3	18.7	22.7	25.5	30.8	33.1	32.0	28.9	25.4	21.7	15.7
	5%	WB	9.5	10.9	14.3	18.5	21.7	25.8	27.8	27.7	25.9	21.9	18.6	12.5
		MCDB	10.8	13.0	16.9	21.6	24.5	29.1	31.9	31.0	28.3	23.6	20.1	13.9
	10%	WB	7.8	9.3	12.9	17.2	20.7	24.8	27.3	27.3	25.2	20.6	17.3	10.9
		MCDB	9.2	10.4	15.5	20.3	23.4	27.5	31.1	30.5	27.5	23.0	19.0	12.6
Mean Daily Temperature Range	5% DB	MDBR	6.1	6.2	7.4	7.8	7.3	5.5	6.4	5.6	5.1	6.1	6.8	6.8
		MCDBR	9.3	10.8	11.4	11.8	10.8	8.7	8.8	7.4	6.2	6.5	8.1	8.7
		MCWBR	6.2	7.1	6.5	6.1	5.0	3.8	3.1	2.9	3.0	3.6	4.9	5.9
	5% WB	MCDBR	7.7	8.8	9.8	9.4	7.9	7.6	7.8	6.4	5.6	5.3	6.1	6.9
		MCWBR	5.8	6.7	6.8	5.8	4.7	3.7	3.3	2.9	3.3	4.0	4.7	5.7
Clear-Sky Solar Irradiance	taub	0.580	0.619	0.656	0.663	0.647	0.698	0.665	0.633	0.586	0.579	0.594	0.571	
	taud	1.705	1.653	1.609	1.651	1.721	1.668	1.769	1.825	1.891	1.813	1.729	1.730	
	Ebn at Noon	611	638	656	677	693	656	675	690	703	664	595	590	
	Edh at Noon	198	228	254	252	237	248	224	209	189	191	192	184	
All-Sky Solar Radiation	RadAvg	2.20	2.62	3.72	4.38	4.74	4.16	5.37	5.04	4.07	3.42	2.47	2.15	
	RadStd	0.28	0.50	0.31	0.49	0.51	0.60	0.64	0.65	0.41	0.44	0.30	0.31	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+89	+59	

Nomenclature: See separate page